arXiv:2402.18344v2 [cs.CL] 27 Jun 2024

Focus on Your Question! Interpreting and Mitigating Toxic CoT Problems in Commonsense Reasoning

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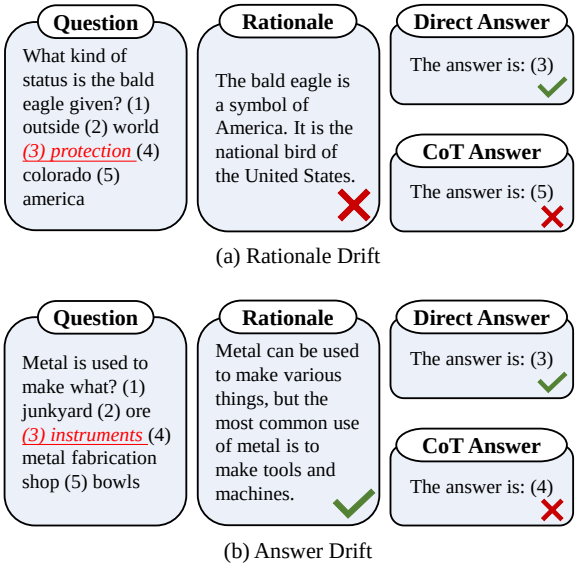
Abstract

Large language models exhibit high-level commonsense reasoning abilities, especially with enhancement methods like Chain-of-Thought (CoT). However, we find these CoT-like methods lead to a considerable number of originally correct answers turning wrong, which we define as the Toxic CoT problem. To interpret and mitigate this problem, we first utilize attribution tracing and causal tracing methods to probe the internal working mechanism of the LLM during CoT reasoning. Through comparisons, we prove that the model exhibits information loss from the question in the shallow attention layers when generating rationales or answers. Based on the probing results, we design a novel method called RIDERS (**R**esidual decod**I**ng and s**ER**ial-position **S**wap), which compensates for the information deficit in the model from both decoding and serial-position perspectives. Through extensive experiments on multiple commonsense reasoning benchmarks, we validate that this method not only significantly eliminates Toxic CoT problems (decreased by **23.6%**), but also effectively improves the model's overall commonsense reasoning performance (increased by **5.5%**).

1 Introduction

With the increase in scale, large language models (LLMs) have demonstrated outstanding performance in different tasks (Li et al., 2023; Wen et al., 2023; Sun et al., 2024; Jin et al., 2024a), among them, commonsense reasoning has received significant attention due to its importance for general intelligence (Wang et al., 2022, 2024; Liu et al., 2024). In this task, researchers have proposed a series of chain-of-thought (CoT) like techniques to elicit models' potential abilities (e.g. Self-Consistency (Wang et al., 2023c), Least-to-Most (Zhou et al., 2023), Reflexion (Shinn et al., 2023)).



(a) Rationale Drift

(b) Answer Drift

Figure 1: Two examples for the Toxic CoT problem.

Through them, LLMs can generate reasonable rationales and improve their reasoning performance.

While these works have made great progress, we notice an overlooked problem in them, which we define as **Toxic CoT** — Sometimes LLMs can directly provide correct answers to questions, but after applying CoT-like methods, it brings extra reasoning paths to models, causing their answers to be wrong.[1] Figure 1 illustrates two main error types of this problem — Rationale Drift and Answer Drift. Specifically, for the Rationale Drift case, given the question "*What kind of status is the bald eagle given?*", the model can directly give the correct answer "*protection*". However, in the rationale, the model explains "*what is the bald eagle*" as "*a symbol of America*", which has a semantic drift from the question. Thus, the model chooses the wrong option "*america*" based on the drifting rationale. For the Answer Drift case, given the question

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[1]Notably, our definition here differs from the toxicity in generated content (Shaikh et al., 2023), emphasizing the harm that CoT can bring to the model.

"*Metal is used to make what?*", the model can directly answer "*instruments*". It can also generate a correct rationale "*metal is to make tools and machines*", but when answering based on the rationale, the model drifts from it and selects the incorrect option "*(4)*". We further conduct a statistical analysis over extensive commonsense reasoning datasets and find that, among all CoT errors, this problem accounts for **37%** for the white-box model and **33%** for the black-box model on average, indicating this problem has become a crucial bottleneck in CoT reasoning.[2]

So what is the mechanism behind this issue? In this paper, we attempt to answer this question by probing the inner workings of the LLM's CoT reasoning. Specifically, we first make initial observations on examples of Rationale Drift and Answer Drift issues, which suggest that the model likely misses some important information from the question when generating corresponding rationales or answers. To further verify these findings, we use attribution tracing and causal tracing methods to probe the LLM in two stages (rationale generation stage and answer generation stage). By employing these methods under various experimental settings, **we find that there is a significant loss of information flow from the question in the shallow attention layers when generating drifting rationales and answers.** Therefore, we interpret the Toxic CoT problem as the model lacking information from the question in the two stages.

To validate our interpretation and mitigate this problem, we design an approach called $\mathbb{RIDERS}$ (**R**esidual decod**I**ng and s**ER**ial-position **S**wap) based on the interpretation. Concretely, for the Rationale Drift issue, we devise a decoding algorithm, promoting the model to generate tokens that pay more attention to question contexts. For the Answer Drift issue, we swap the positions of the output sequence, reducing the information loss from the question to the final prediction. We evaluate our method on five commonsense reasoning benchmarks and conduct extensive experiments. The results not only prove our interpretation, but also indicate that our method is effective in addressing the Toxic CoT problem and improving the model's overall commonsense reasoning abilities.

We summarize the contribution of this paper as follows:

(1) We identify a crucial bottleneck affecting LLM's reasoning performance called the Toxic CoT problem, probe this issue through attribution tracing and causal tracing methods, and interpret the mechanism behind it as the model missing information from questions in shallow attention layers. The results contribute to a more in-depth understanding of the LLM's reasoning mechanisms.

(2) To mitigate the Toxic CoT problem, we introduce $\mathbb{RIDERS}$, which effectively compensates for the internal information loss during CoT reasoning from decoding and serial-position perspectives.

(3) We conduct extensive experiments on various benchmarks. The results not only verify the rationality of our interpretation, but also demonstrate the effectiveness of our method in addressing the Toxic CoT problem (the proportion of the problem decreased by **23.6%**) and enhancing commonsense reasoning performance (overall accuracy increased by **5.5%**). Our code is available at: https://github.com/BugMakerzzz/toxic_cot.

2 Problem Statement

2.1 Toxic Chain of Thought Reasoning

We start our work by formally defining the Toxic CoT problem as follows: [3]

Definition 2.1 (Toxic CoT). *Given a question q and the correct answer o^*, if the model's output $\mathcal{M}$ meets the following conditions, it is considered a case of Toxic CoT:*

$$o^* = \mathcal{M}(q, P_d) \wedge o^* \neq \mathcal{M}(q, P_c)$$

where $o^* = \mathcal{M}(q, P_d)$ indicates the model's direct answering for q is correct, $o^* \neq \mathcal{M}(q, P_c)$ indicates the model's cot-like answering for q is wrong, P_d, P_c are the corresponding prompts.

2.2 Two-stage Drift Issues

To investigate the reasons for the problem, we classify these Toxic CoT cases and identify a main error causing this problem (On average, it accounts for 67% on two datasets, see more details in Appendix B). Furthermore, if we divide the CoT process into two stages: **rationale generation** and **answer generation**, there exist two types of issues in this error:

Definition 2.2 (Rationale Drift). *If the reasoning chain is factually correct but logically inconsistent*

[2] Appendix A presents the details of settings and results in this statistical experiment.

[3] In practice, CoT-type methods all have Toxic CoT problems, but to simplify the work, this paper mainly focuses on the basic CoT prompting.

with the question, this case is called "Rationale Drift" (see Figure 1a).

Definition 2.3 (Answer Drift). *If the reasoning chain is both factually correct and logically consistent with the question, but the final answer is inconsistent with the rationale, this case is called "Answer Drift" (see Figure 1b).*

2.3 Hypothesis Formulation

To provide a direction for subsequent probing experiments, here we attempt to propose hypotheses for the mechanism of issues by analyzing some examples. For the Rationale Drift issue, the model tends to focus on part of the essential reasoning conditions in the question context. As an example, in Figure 1a, the CoT only focuses on the "*bald eagle*" in the question but misses another key information "*status*". As for the Answer Drift issue, the model seems to be disrupted by CoT, losing attention to the question and resulting in an off-topic prediction. For instance, in Figure 1b, though the CoT gives correct information "*to make tools and machines*", the model can only predict the wrong answer "*(4) metal fabrication shop*". This is likely because the model loses the question's target "*to make what*" and directly copies the entity "*metal*", which frequently appears in CoT, as the answer. Therefore, we summarize our hypotheses as follows:

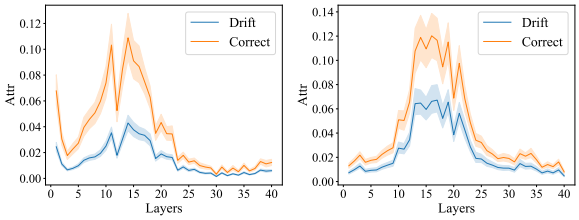
Hypothesis 1. *The Rationale Drift issue arises from the model lacking information from the question context in the rationale generation stage.*

Hypothesis 2. *The Answer Drift issue arises from the model lacking information from the question in the answer generation stage.*

To validate the above hypotheses, in the following two sections, we conduct probing experiments, exploring the LLM's internal working mechanisms during the two stages of CoT reasoning.

3 Tracing Information Flow in Rationale

In this section, we aim to verify the Hypothesis 1 by tracing the information flow in the rationale generation stage. To this end, we start by describing our attribution tracing method (§3.1). Through this method, we conduct comparative experiments between the correct reasoning and the drifting one, figuring out the mechanism behind the issue (§3.2). At last, we use the attention score to validate our findings from another perspective (§3.3).



(a) Llama2-13B (b) Baichuan2-13B

Figure 2: Attribution tracing results on Winogrande.

3.1 Tracing Method

To investigate the roles of different model components in the rationale generation stage, we use attribution scores (Hao et al., 2021; Dai et al., 2022; Wang et al., 2023b) to compute the contribution of a neuron ω:

$$Attr(\omega) = \omega \odot \int_{\alpha=0}^{1} \frac{\partial F(\alpha\omega)}{\partial \omega} d\alpha \approx \frac{\omega}{m} \odot \sum_{k=1}^{m} \frac{\partial F(\frac{k}{m}\omega)}{\partial \omega} \tag{1}$$

where $F(\cdot)$ represents the model's output. We compute the attribution score via Riemman approximation of the integration and m is the number of approximation steps. For neurons $A^{(l)}$ in the i-th attention layer, we sum the absolute values of scores on all attention heads to get the final attribution score. Since the attention module involves interactions between different tokens, we can compute the information flow between the question context q and the CoT c on it:

$$\begin{aligned} Q_{qc}^{(l)} &= \frac{1}{|N|} \sum_{(i,j)\in C_{qc}} Attr(A_{i,j}^{(l)}) \\ C_{qc} &= \{(i,j) | q_s \le i \le q_e,\ c_s \le j \le c_e\} \end{aligned} \tag{2}$$

Here, $Attr(A_{i,j}^{(l)})$ represents the intensity of information flow from the i-th token to the j-th token in the l-th attention layer and $|N|$ denotes the number of CoT steps. More implementation details of this method are reported in Appendix C.

3.2 Attribution Tracing Experiment

Experimental Settings To validate the deficiency in Hypothesis 1, we need to figure out the context information flow difference between generating a drifting rationale and a correct one. Thus, we first use golden labels as hints to generate correct CoTs in the drifting cases. Then, we compute the average information flow under these two cases and compare their results. We choose Llama2-13B (Touvron et al., 2023) and Baichuan2-13B (Yang et al., 2023) as our probing models, since they are moderated-sized white-box models with decent